

7 : Suppose A , B and C are sets. If $B \subseteq C$, then $A \times B \subseteq A \times C$.

*Consider $\forall x \in B$, those elements x are also in C .
Then every element in set $\{A, B\}$ would be in set $\{A, C\}$.
Thus $A \times B \subseteq A \times C$.*

12 : If A , B , and C are sets, then $A - (B \cap C) = (A - B) \cup (A - C)$

Let's evaluate both sides.

$A - (B \cap C)$ means element x is in A and not in $B \cap C$.

*LH Since $B \cap C$, would've meant element x is in B and C ,
element $A(x)$ can't be in B or C .*

So the left-hand side states $\forall x \in A, x \notin B, C$.

$(A - B) \cup (A - C)$ starts as element x is in $A - B$ or $A - C$:

$A - B$ means element x is in A but not B .

RH $A - C$ means element x is in A but not C .

These are all true only if element x is in A and not in B or C .

Symbolically, $\forall x \in A, x \notin B, C$.

Bringing it together : $(\forall x \in A, x \notin B, C) = (\forall x \in A, x \notin B, C)$

So this statement is true.

18 : If A, B and C are sets, then $A \times (B - C) = (A \times B) - (A \times C)$.

Let's evaluate.

$$\left| \begin{array}{lcl} A \times (B - C) & = & (A \times B) - (A \times C) \\ (A \times B) - (A \times C) & = & (A \times B) - (A \times C) \end{array} \right|$$

*Since $A \times (B - C)$ can be written as $(A \times B) - (A \times C)$,
it is the same and thus true.*

22 : Let A and B be sets. Prove that $A \subseteq B$ if and only if $A \cap B = A$

Let's consider the conditions.

Case 1 : If $A \subseteq B$ then $A \cap B = A$

Proof : $\left\{ \begin{array}{l} A \subseteq B \rightarrow \text{if } \forall x \in A, \text{ then } x \in B \\ A \cap B \rightarrow \forall x, x \in A \cap B \end{array} \right.$
 When the elements in A are a subset of B , only the elements in A satisfy both of these. So $A \cap B$ can only equal A .

Case 2 : If $A \cap B = A$, then $A \subseteq B$

Proof : $\left\{ \begin{array}{l} A \cap B = A \rightarrow \text{element } x \text{ is in } A \text{ and } B, \text{ thus } \forall x \in (A \cap B), x \in A. \\ \text{If all of the elements that are in } A \text{ are also in } B, \text{ then } A \text{ would be a subset of } B. \end{array} \right.$
 This statement proves true by the definition of sets.

Thus, $A \subseteq B$ if and only if $A \cap B = A$ is true.

29 : Suppose $A \neq \emptyset$. Prove that $A \times B \subseteq A \times C$ if and only if $B \subseteq C$.

Consider the two statements :

Statement 1 :

if $A \times B \subseteq A \times C$, then $B \subseteq C$
 $A \times B$ makes a new set with combinations of $\{x, y\}$;

$\{x, y\} \left\{ \begin{array}{l} \{x_{n-(n-1)}, y_{p-(p-1)}\} \\ \{x_{n-(n-1)}, y_{p-(p-2)}\} \\ \{x_n, y_p\} \end{array} \right.$

Where

$|A| = n, |B| = p,$

$\forall x \in A, \forall y \in B.$

Since A represents x on both sides of the statement, it is the same set, or $A \subseteq A$.

For y that is B on the left side and C on the right.

Therefore, all elements $y \in B$ are also in C . Thus $B \subseteq C$.

Statement 2 :

if $B \subseteq C$, then $A \times B \subseteq A \times C$
 Consider that for $B \subseteq C$, that means :

if $\forall y \in B$, then $\forall y \in C$.

Consequently, for the cross product pair :

$\{x, y\}$ in $A \times B \subseteq A \times C$,

if $x \in A$, then $x \in A$ for $A \subseteq A$,

if $y \in B$, then $y \in C$ for $B \subseteq C$.

Therefore, this statement is true.